

SOCIAL MEDIA SENTIMENT ANALYSIS

Airline Customer Experience & Service Conversations

Professional Internship Project Report

Metric	Value
Tweets analyzed	14,485
Negative	9,082 (62.7%)
Neutral	3,069 (21.2%)
Positive	2,334 (16.1%)
Airlines	6

1. Objective

Analyze social-media conversations using sentiment labels and NLP techniques; visualize sentiment distribution, compare airlines, examine sentiment over time, identify recurring negative reasons, and extract common keywords.

2. Dataset & Scope

The supplied Twitter Airline Sentiment dataset contains 14,640 tweets from February 2015 covering six airlines. It includes tweet text, airline, sentiment, negative reason, date/time, and engagement fields.

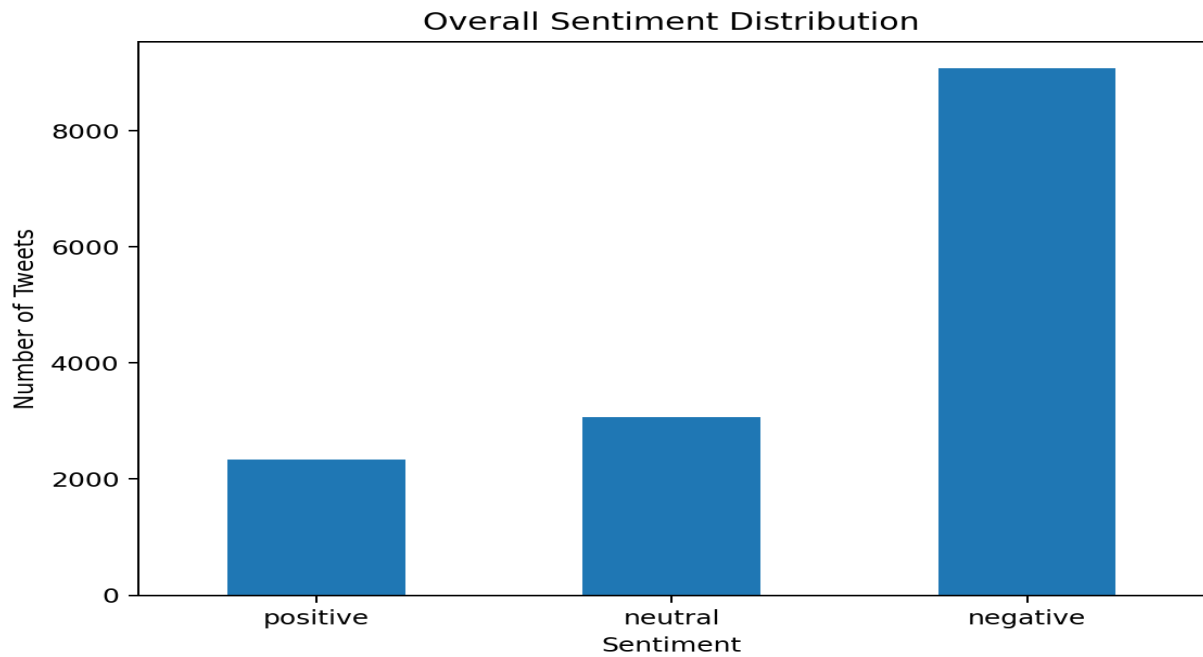
Scope limitation: the source data is historical, so results describe the available dataset period rather than current 2026 public opinion.

3. Methodology

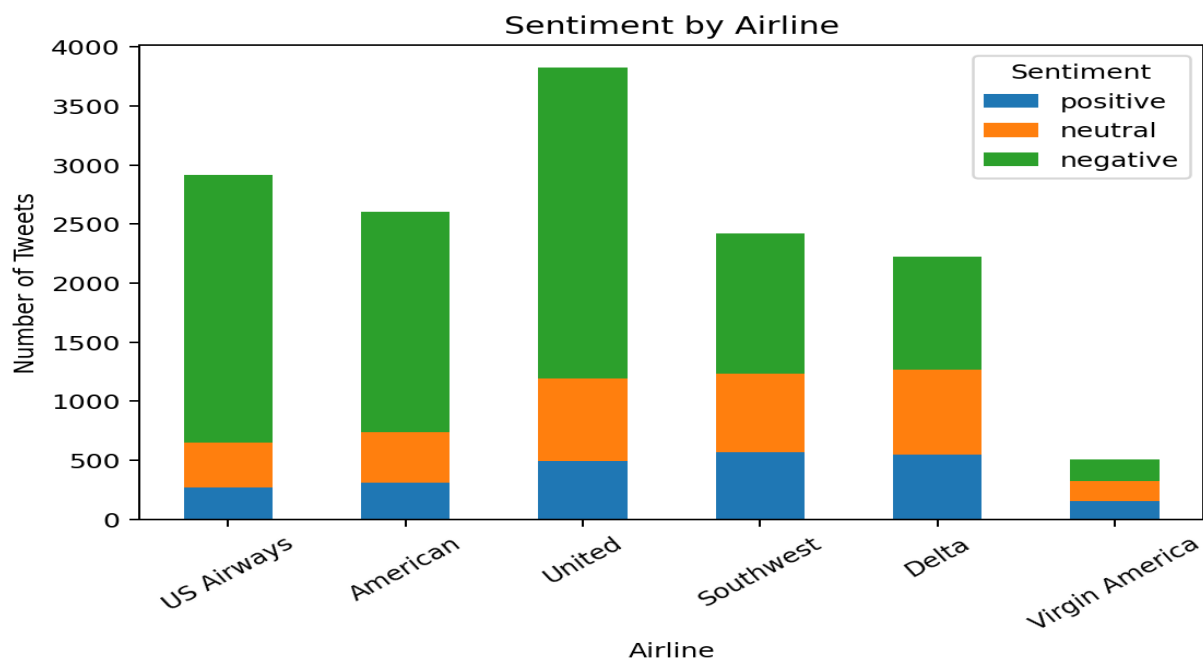
Duplicates are removed by tweet ID, timestamps are parsed, text is normalized for keyword analysis, the supplied labels are preserved, and VADER is included as a secondary NLP check. Matplotlib visualizes the results and WordCloud summarizes recurring terms.

4. Key Findings

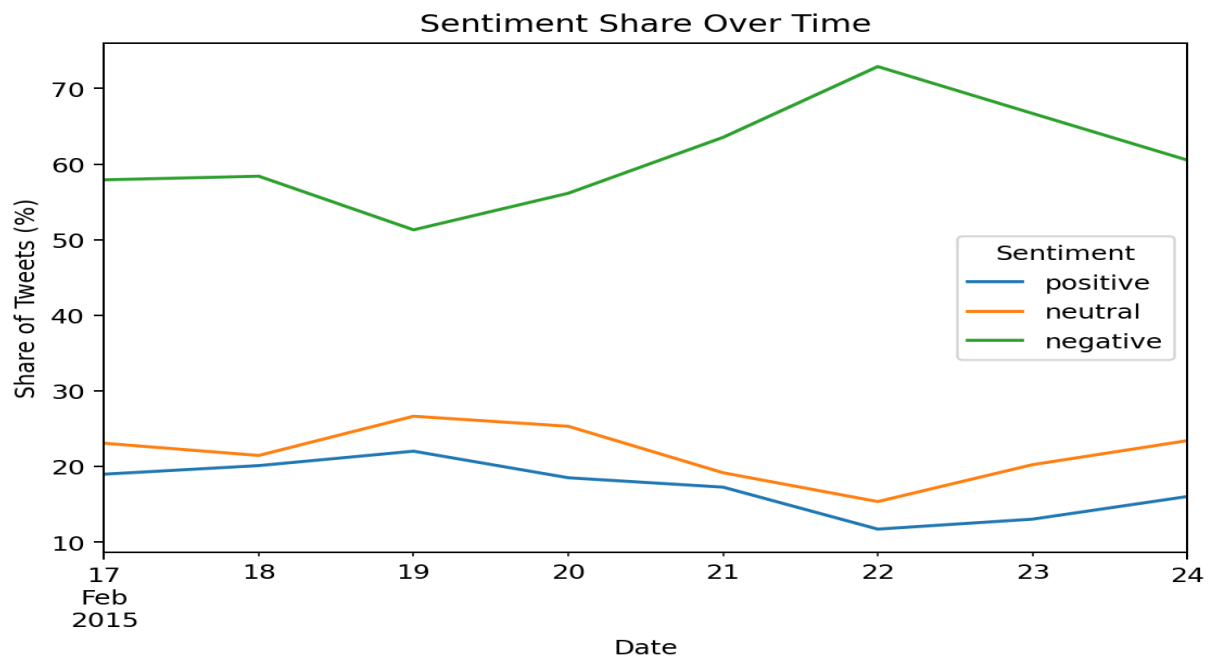
Negative sentiment represents 62.7% of the dataset, neutral 21.2%, and positive 16.1%. The highest negative-share airline in this dataset is **US Airways**. The most frequent recorded negative reason is **Customer Service Issue**.



Overall sentiment distribution



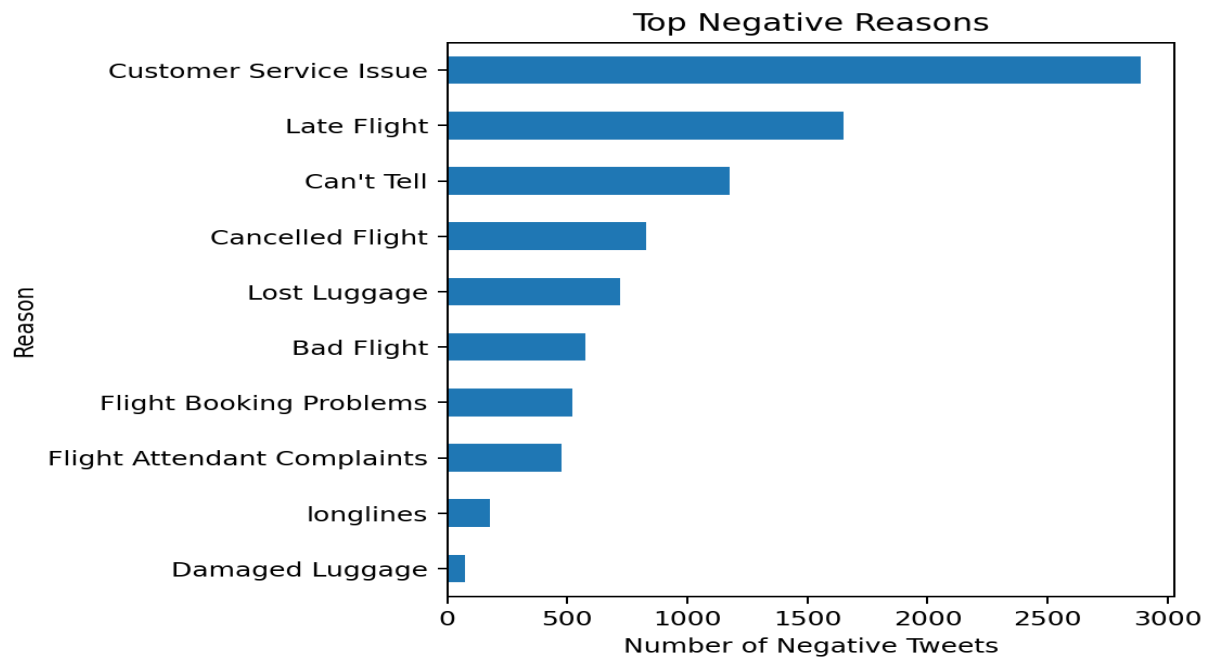
Sentiment comparison by airline



Sentiment share over time

5. Negative Issues & Keywords

The negative-reason chart highlights recurring service issues, while the word cloud summarizes frequent terms in negative conversations.



Top negative reasons



Negative tweet keyword word cloud

6. Conclusion

This submission demonstrates an end-to-end social-media analytics workflow: cleaning, sentiment analysis, comparison, time analysis, issue discovery, keyword extraction, visualization, and communication of findings.